

ADEWOLE HABEEB ADEBOLA

Software Engineer · BSc Computer Science (Final Year)

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ABOUT ME

Final-year Computer Science student and software developer with hands-on experience building applications end to end, from React and TypeScript on the frontend to Python and FastAPI on the backend, plus machine learning for my final year research on financial fraud detection. As President of my university's NACOS chapter for the 2025/2026 session I led a full executive administration, delivering the department's first hackathon and gaming festival as well as three digital systems that replaced manual processes. I am comfortable owning a feature from design handoff through to deployment, and I write the documentation to match.

PROFESSIONAL EXPERIENCE

President | NACOS, Crescent University Abeokuta Chapter
2025 – 2026

- Led the chapter executive administration for the 2025/2026 session, setting priorities across technical, welfare, academic and social portfolios for the Computer Science department.
- Launched two first-ever departmental events, the NACOS CUAB Hackathon and NACOS GameFest, establishing both as fixtures on the departmental calendar.
- Secured and delivered industry-facing workshops in partnership with Zulfah Group and the Ogun Tech Community, the latter focused on AI and data.
- Drove the chapter's digital transformation: commissioned a complete website redesign, an Electronic Voting System for chapter elections, and a Departmental Receipt System that digitised dues collection and record keeping.
- Rebuilt and revamped the chapter Dev Team, growing the technical community and creating hands-on build opportunities for students across year groups.
- Delivered the department's social calendar including Dinner and Awards Night, Movie Night and Sports Day, handling nomination data cleaning, de-duplication of nominee records and results integration from the official election portal.
- Oversaw department-wide student welfare, acting as the primary channel between the student body and faculty leadership.

Frontend Developer | Latex Fabrics, Lagos (Freelance)
2025

- Engaged by a Lagos-based fabric retailer to scope and design a progressive web application for online catalogue browsing and ordering.
- Delivered the full pre-build package: brand documentation, product requirements document, functional requirements document, design system and wireframe specifications.
- Translated a non-technical client's commercial goals into concrete, buildable product requirements.

Frontend & Java Developer Intern | APTECH Computer Education, Nigeria
2023

- Developed Java applications applying object-oriented principles including inheritance, static members and method overriding.
- Used Java's Regular Expression API (Pattern and Matcher) for text parsing and validation routines.
- Built self-contained projects including a banking application and a student grading system.
- Implemented core frontend interfaces using HTML, CSS and JavaScript.

Peer Tutor | Crescent University Abeokuta
2023 – Present

- Provide academic support to fellow students in Computational Science, Numerical Methods, Algorithms & Complexity and Discrete Structures.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, Python, HTML, CSS, C++ (basic)
Frontend: React 18, Next.js 14, Vite, Tailwind CSS, responsive and accessible UI
Backend & Data: Flask, FastAPI, MySQL, Supabase, REST API design, JWT auth
Machine Learning: scikit-learn, PyTorch, PyTorch Geometric, pandas, NumPy, imbalanced-learn
Tools: Git & GitHub, pytest, Figma, Playwright, VS Code, IntelliJ IDEA

PERSONAL SKILLS

Clear written and verbal communication
Leadership and delegation
Stakeholder and vendor liaison
Critical and creative problem solving
Time management under deadline
Technical documentation

CERTIFICATIONS

Java Programming Certificate, APTECH Computer Education, Nigeria (2025)
Certificate of Appointment as Elected President, NACOS CUAB Chapter (2025)

LEADERSHIP

President, NACOS Crescent University Abeokuta Chapter (2025/2026 session)

REFERENCES

Available upon request

- Raised peer performance in core theory modules while reinforcing personal mastery of the material.

EDUCATION

Crescent University, Abeokuta | BSc Computer Science

Expected graduation 2026

Cumulative GPA 4.27 / 5.0, Second Class Honours (Upper Division). Final results pending. Major GPA 5.0 / 5.0.

PROJECTS

Tracer | Final Year Project

- Built an end-to-end machine learning system detecting illicit Bitcoin transactions, achieving 0.807 F1 on the illicit class (92.5% precision, 71.6% recall, 0.944 AUC-ROC) and reproducing the published benchmark of Weber et al. (2019) on the same dataset and evaluation protocol.
- Trained and compared three models (Random Forest, a GraphSAGE graph neural network, and an RF/GNN-embedding hybrid) on the Elliptic dataset of 203,769 Bitcoin transactions, 234,355 payment-flow edges and 165 features, applying SMOTE for class rebalancing and a leakage-free temporal train/test split to prevent future data leaking into training.
- Delivered the models through a three-tier web application: a React 18, Vite and Tailwind front end, a Flask REST API with JWT authentication and server-side role-based access control across six blueprints, and an 8-table MySQL schema persisting the transaction graph.
- Implemented continuous stream monitoring that automatically screens every transaction per time step and generates analyst alerts under a per-step capacity budget, alongside a 2-hop network visualisation of each scored transaction's payment neighbourhood.
- Built per-prediction explainability by implementing decision-path (Saabas) attribution in NumPy over scikit-learn's tree internals after SHAP proved unusable in the target environment; verified exact additivity and overall system behaviour with a 35-test pytest suite.

VectoGen | Text-to-Vector Graphics Platform

- Platform generating editable vector graphics from natural language prompts, built without reliance on third-party AI APIs.
- Three self-built components: an NLP intent parser (scikit-learn), a programmatic SVG assembly engine, and a design variation system.
- Served via a FastAPI microservice to a Next.js 14 frontend, with Supabase handling authentication and storage.

Departmental Receipt System (DRS) | Frontend

- Receipt and payment-record system built to digitise departmental dues collection for the NACOS chapter, replacing paper records.
- Frontend built in React and TypeScript directly against Figma designs via a Figma MCP integration.

Opportuna | Startup Project

- Co-founded a student-focused internship platform connecting students with verified internship opportunities while streamlining employer recruitment workflows.
- Led frontend development using React, TypeScript and Tailwind CSS, building responsive, accessible interfaces for students, employers and administrators.
- Designed and implemented core product flows including authentication, internship discovery, application management, profile creation and role-based dashboards.
- Collaborated on product requirements, design system and API integration, translating Figma designs into production-ready components while maintaining consistent UI/UX across the application.
- Worked closely with the team from product ideation through MVP development, helping shape the platform architecture and user experience.